Data Models (Module-II)

Dr.Hemalatha K
Assistant professor
SCOPE,VIT Chennai.

What is IoT Data management

It enables users to **track, monitor and manage the devices to ensure the devices to work properly and securely after deployment.**

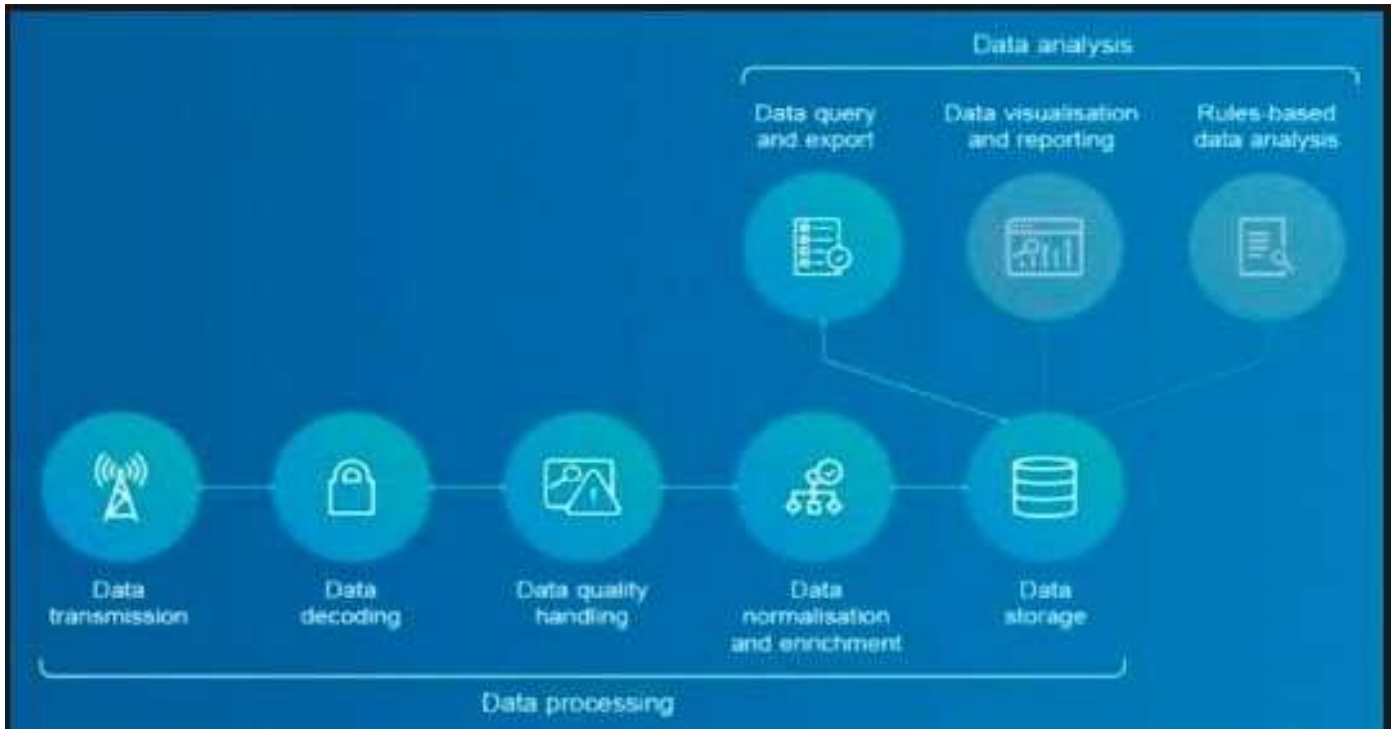
Why Data Management is need?

- **IoT sensors** interact with people, homes, cities, farms, factories, workplaces, vehicles, wearables and medical devices, and beyond.
- it is changing our lives from managing home appliances to vehicles. The technology of devices can now advise us about w**hat to do, when to do** and **where to go**.
- **Industrial IoT applications assist us in managing the data for processes, and predicting faults and disasters.**
  - The IoT platforms help set and maintain parameters to refine and store data accordingly.

Data Management Process

- The process of taking the overall available data and refining it down **to important information.**
  - Different devices from different applications send large volumes and varieties of information. Managing all this IoT data means is **developing and executing architectures, policies, practices and procedures that can meet the full data lifecycle needs**.
- Things are controlled by smart devices to automate tasks, so we can save our time.

  Intelligent things can collect, transmit and understand information, but a **tool** will be **required to aggregate data** and **draw out inferences, trends and patterns**.



IoT data management techniques

cloud computing

- **data processing happens in a centralised data storage location**
- Sensors and devices can connect indirectly through the cloud, where data is centrally-managed, or send data directly to other devices to locally collect, store and analyse the data, and then share selected findings or information with the cloud

Edge computing

- **data is processed near the data source or at the edge of the network.** By processing and using some data locally, the IoT saves storage space for data, processes information faster and meets security challenges.
- Sensors produce a large amount of data for edge gateway devices so that these can make decisions by analysing the data
- Edge devices for data management help secure the most **valuable data and reduce bandwidth cost**.

Data Management challenges

- **Space Optimization :** the number of IoT devices will increase, thus increasing the challenges for **real-time processing and analysis to reduce time for storage.**
- **Identification of tools**: Functions such as **adaptive maintenance, predictive repair, security monitoring and process optimization** rely on real-time data selecting the right tools is a **challenge because integration between different sensors should be proven and compatibilities confirmed.**
- When there is no connection, devices must still gain insights, make decisions and prepare for data distribution.

Tagging of Data

Data forms

- Text data on paper
- Bytes as electronic memory storage.
- Twitter data
- image data
- structured data
- semi structured data
- unstructured data
- views
- likes
- tagged data, meta data

Example-I

- Google Data power
  - Through an email can seen a pop-up ad, that you saw from the browser before.
  - How it is happening?
    - Google software read your email, saw that you mentioned unicorn hoodies, found an advertisement for one, and then offered them to you.
    - Google collects billions of data points every single second from users around the world.
    - Data points come from Gmail, Google searches, G Suite, Google Chrome, YouTube,

Example -II

- Real estate website that provides home values for free.
- They started making agreements with local real estate associations to port their local sales information directly to Zillow's database.
- Zillow gets data from realtors. It uses that data to attract homebuyers online, generating leads for realtors, and then sells this data back to the realtors, who gave them the data in the first place.

Data organization

- Data is more powerful when it is organized properly
  - It's the ability, as a business owner, to see and **understands the trends in your customers' behavior** and make **proactive, rather than reactive** changes in order to increase your bottom line sales

Advantages of Data collectors

- Monetization of data for **gaining the money**.
  - Social media networks such as Facebook, Twitter, and LinkedIn use data to make money.
- Collecting data on your customers and their trends

Tags



- Organizing the data
  - Collecting data
  - Learn how to track
  - Using the data
- Tags are something you can **assign to any set of data**. A set of data can be tagged with one, or multiple tags.
  - social media post with hashtags.
- Process of Tagging
  - is a kind of indexing, a **process** of labelling and categorizing information made to support resource discovery for users.
    - Social **tagging** generally means the practice whereby internet users generate keywords to describe, categorize or comment on digital content
- *Tags* are a very efficient way to organize data sets, such as customers,
  - improve the experience businesses with categoric data
- Some data with multiple tags.
  - Customer might be with more than one tag.
- **CRMs have multiple data fields that you can input for each customer. Any field where you can add a category that is common across the different customers, and if you can categorize them with more than one of those category options, using tags.**

HashTag



- Assigning the set of data of social media data with hashtags.
- A **hashtag** is a metadata **tag** that is prefaced by the **hash** symbol, #. Hashtags are widely **used on microblogging and photo-sharing services** such as **Twitter and Instagram** as a form of user-generated **tagging** that enables cross-referencing of content sharing a subject or theme.
- hashtag is used to draw attention, organize, promote, and connect. It is started on Twitter as a way of making it easier for **people to find, follow, and contribute to a conversation**
- Instagram every post is tagged with a specific hashtag.
  - a hashtag called **#chicagofood** that is attached to about 791,000 posts.
  - #ginoseastpizza (a restaurant in Chicago) which is attached to just over 1000 posts

Why use hashtags?

- **Hashtags helps to get found of target audience.**
- **Hashtags improve your click through rates (CTR)**
- **Hashtags are great for research**
- **Hashtags bring like-minded people together.**
- **Hashtags can be used for humor**

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- **#business**, **#entrepreneur** or **#marketing**
- **Specifying**
  - put the pound sign directly in front of the word or phrase you want to turn into a hashtag and follow these simple rules:
  - No spaces
  - No punctuation
  - No special characters
  - capitalization for readability(optional)

WITHOUT THE TAGS, SENDING EVERYONE IN YOUR DATABASE EVERYTHING YOU HAVE TO OFFER, WHICH IS NOT AS EFFECTIVE.

Types of Tags

- One Tag
- Multiple Tags
- Hyper-targeted advertising tags

One Tag

- Only one tag is associated with the data of the customer data
  - Buying only beauty products by the customer
  - Buying only sports goods from the same company

Multiple tags

- Assigning more than one tag to a customer by the company
  - the customer who buys beauty products and sporting goods
  - **With tags, a company can assign as many different tags to a customer as they need and desire.**
  - Now it's up to the company's marketing team to market to customers based on how they are tagged;

Targeted advertising

- Targeted advertising is a form of advertising, an online advertising, that is directed towards an audience with **certain traits(quality parameters)**, based on the product or person the advertiser is promoting.
- These traits can either be demographic with a focus on race, economic status, gender, age, generation, level of education, income level, and employment, or there can be a psychographic focus which is based on the consumer values, personality, attitude, opinion, lifestyle and interest.

Importance of target advertising

- Digital data radically increase, and it spreads across numerous ICT channels cohesively.
- The emergence of new online channels, the **need for targeted advertising is increasing**
- Companies aim to minimize wasted advertising by means of information technology.

Issues with targeted advertising

- the lack of 'new' advertisements of goods or services.
  - Seeing as **all ads are tailored to be based on user preferences**, no different products will be introduced to the consumer.

– In this case the consumer will be at a loss as they are not exposed to anything new.

Hyper-targeted advertising

- Hyper-targeting is a marketing strategy where you clearly **identify a target customer and deliver extremely relevant messages in the** places where they will be most likely to see it.
- Definitions in Hyper-targeted advertising
  - Buyer persona
    - More details about customer **profile outlines who your target customer** is by defining their demographics, sociographic, professional roles, values, goals, challenges, influences and buying habits
  - Segmentation
    - Dividing **the audience into smaller groups based on their interactions with your brand or their demographics, socio**graphic, professional roles, values, goals, challenges, influences and buying habits.
  - Geo-targeting
    - The act of targeting **a customer based on their location**. This could be as broad as targeting by country or as narrow as targeting by zip-code or mile-radius around a location or business.
  - Retargeting
    - The **act of sending messages to audiences who have already engaged with your brand**. This could include remarketing to people who have visited your website, visited your store, made a purchase, followed your brand on social, signed up for an online based offer, or had any other interaction with your brand.

Benefits

- Deliver personalized marketing **messages that are relevant to what customers actually what and need**, and stop sending customers marketing messages about products and services they **have no interest in**.
- Get a better value for your ad spend because you **can reach the right audiences and not spend money on** ad campaigns that are reaching the wrong audiences.
- Get the qualified leads from people who are likely to become customers, and **avoid filling your funnel or pipeline with uninterested audiences.**
- Generate more sales by **marketing the right offerings to the right customers on the right platforms**.
- When executed properly, hypertargeting can make it **easier for you to connect with customers, generate leads, and increase** your sales.

Scenario

- Consider a situation where shop keeper wants to know which products are needs more to buy for his next seasons.
  - Who is his regular customers
  - Which items are seasonal based.
  - Ordering the stock based on predictions.
- Detail about the data collection process for the above problem and how the data will be organized from it and details any technique that will structure the data.

Tag data to pre-process large datasets

Steps to Tag Data for Pre-Processing Large Datasets

1. **Understand the Dataset**:
   - Analyze the dataset to identify important **features, patterns, or categories** that need tagging.
   - Example: In a dataset of customer reviews, tags could include "Positive Sentiment," "Negative Sentiment," or "Neutral Sentiment."

2. Define Tagging Criteria:

- Determine the **rules for creating and applying** tags.
- Tags could be based on:
  - Categories: "High Priority," "Low Priority."
  - Context: "E-commerce," "Healthcare."
  - Data Type: "Numeric," "Categorical," "Text."

3. Select or Create Tags:

- Use predefined tags or create new ones based on domain-specific requirements.
- Example: For a product catalog, tags might include "Electronics," "Clothing," "Under $50."

4. Automate Tagging:

- Use tools or scripts to automate the tagging process for large datasets.
- Methods include:
  - **Keyword Matching:** Applying tags based on specific words or phrases.
  - **Regular Expressions:** For pattern-based tagging (e.g., detecting email addresses or phone numbers).
  - **Machine Learning Models:** Use classification models to predict and assign tags (e.g., spam detection).

5. Validate Tags:

- Check the accuracy of the tags to ensure they align with the dataset's intended use.
- Example: Verify that all "Spam" emails are correctly tagged in an email dataset.

6. Store Tags Efficiently:

- Include tags as additional columns in structured datasets (e.g., CSV, SQL tables).
- Use metadata fields in unstructured data formats (e.g., JSON, XML).

7. Use Tags in Pre-Processing:

- **Filter Data:** Extract subsets of data based on tags (e.g., "High Priority" rows).
- **Normalize Data:** Use tags to identify columns requiring normalization or standardization.
- **Handle Missing Values:** Tags like "Missing" or "Null" can help locate and impute missing data.

Examples of Tagging Data for Pre-Processing

1. Text Data:

Tag: "Sentiment" → Values: "Positive," "Negative," "Neutral."

Tag: "Topic" → Values: "Sports," "Politics," "Technology."

2. Image Data:

Tag: "Object Detected" → Values: "Cat," "Dog," "Car."

Tag: "Resolution" → Values: "Low," "Medium," "High."

Advantages of Tagging for Pre-Processing

1. **Efficient Filtering:**
   - Tags make it easy to subset and analyze specific portions of the dataset.
2. **Improved Data Quality:**
   - Helps identify outliers, missing values, or errors for correction.
3. **Faster Processing:**
   - Pre-tagged data reduces the time needed for downstream processing.
4. **Enhanced Machine Learning Pipelines:**
   - Tagged data simplifies feature selection and improves model accuracy.

Challenges

1. **Scalability:**
   - Automating tagging for extremely large datasets can be computationally expensive.
2. **Consistency:**
   - Ensuring uniform application of tags across the dataset is critical.
3. **Tag Clutter:**
   - Too many tags or irrelevant tags can complicate the data structure.

Example

- Sample IoT Sensor D

Timestamp	SensorID	(°C)	(%)	Location
2025-01-01 08:00:00	Sensor001	22.5	45	Warehouse1
2025-01-01 08:05:00	Sensor002	23.0	50	Warehouse1
2025-01-01 08:10:00	Sensor003	21.8	48	Warehouse2
2025-01-01 08:15:00	Sensor004	24.2	52	ZoneA
2025-01-01 08:20:00	Sensor001	22.1	46	Warehouse1

- Define Tagging R

Tag	Description
`Location`	Describes where the data was collected (e.g., Warehouse1, ZoneA).
`DeviceType`	Identifies the type of device (e.g., TemperatureSensor).

contd...

- Tagged d

Timestamp	SensorID	Temperature (°C)	Humidity (%)	Location	Tags
2025-01-01 08:00:00	Sensor001	22.5	45	Warehouse1	Location:Warehouse1, DeviceType:TemperatureSensor
2025-01-01 08:05:00	Sensor002	23.0	50	Warehouse1	Location:Warehouse1, DeviceType:TemperatureSensor
2025-01-01 08:10:00	Sensor003	21.8	48	Warehouse2	Location:Warehouse2, DeviceType:TemperatureSensor
2025-01-01 08:15:00	Sensor004	24.2	52	ZoneA	Location:ZoneA, DeviceType:TemperatureSensor
2025-01-01 08:20:00	Sensor001	22.1	46	Warehouse1	Location:Warehouse1, DeviceType:TemperatureSensor

- Filtered D

Timestamp	SensorID	Temperature (°C)	Humidity (%)	Location	Tags
2025-01-01 08:00:00	Sensor001	22.5	45	Warehouse1	Location:Warehouse1, DeviceType:TemperatureSensor
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2025-01-01 08:20:00	Sensor001	22.1	46	Warehouse1	Location:Warehouse1, DeviceType:TemperatureSensor

Contd.,

- Generate CSV for Tagged data
- Analyze the Tagged Data
  - Count the number of records for each location
  - Calculate the average temperature for each location

Data usage

Forecasting

- Ancestors observed the sky to forecast the weather
- Data Scientists develop and train machine learning models to predict sales, risks, events, trends, etc
  - Based on data decisions need to be made about what should be forecast

Forecasting types

- Based on data availability
  - **Qualitative Forecasts** :If there are no data available, or if the data available are not relevant to the forecasts.
    - **Qualitative forecasting** involves using **expert opinions, market research, and judgment** rather than historical data to make predictions. These methods are typically used when **data is limited or unavailable** and rely on **human intuition and experience** to make future projections.
    - Qualitative forecasts are particularly useful for **new products**, **technological innovations**, or in situations where **rapid changes** make historical data unreliable
  - **Quantitative Forecasts** (Time Series Forecasting): If numerical information about the past is available; and the past patterns will continue into the future

Forecasting types

- Based on time horizon
  - Short term forecasts : a normal range between one and three months
    - Use cases : Traffic prediction, stock price prediction, energy demand forecasting, sales forecasting.
  - Medium term forecasts : the time period is normally one year
    - Usecases : Retail,Manufacturing,Energy,Healthcare and Finance.
  - Long term forecasts: predict results over periods greater than two years
  - Usecases : Retail,Manufacturing,Energy,Healthcare and Finance

Techniques: Time series Models (ARIMA,SARIMA,Holt-Winters);Machine Learning Models (Random Forest,XGBoost,LSTM),Exponential Smoothing,Trend Prediction and Regression models etc.,

Forecasting Methods

- **Judgmental Forecasting** is the only option for Qualitative Forecasts due to the lack of historical data ; Examples of a new policy, a new product, or a new competitor
- **Time Series Forecasting** methods only use historical values on the variable to be forecast and exclude the factors that could affect its behavior such as competitor activities, changes in environment or economic conditions, and so on

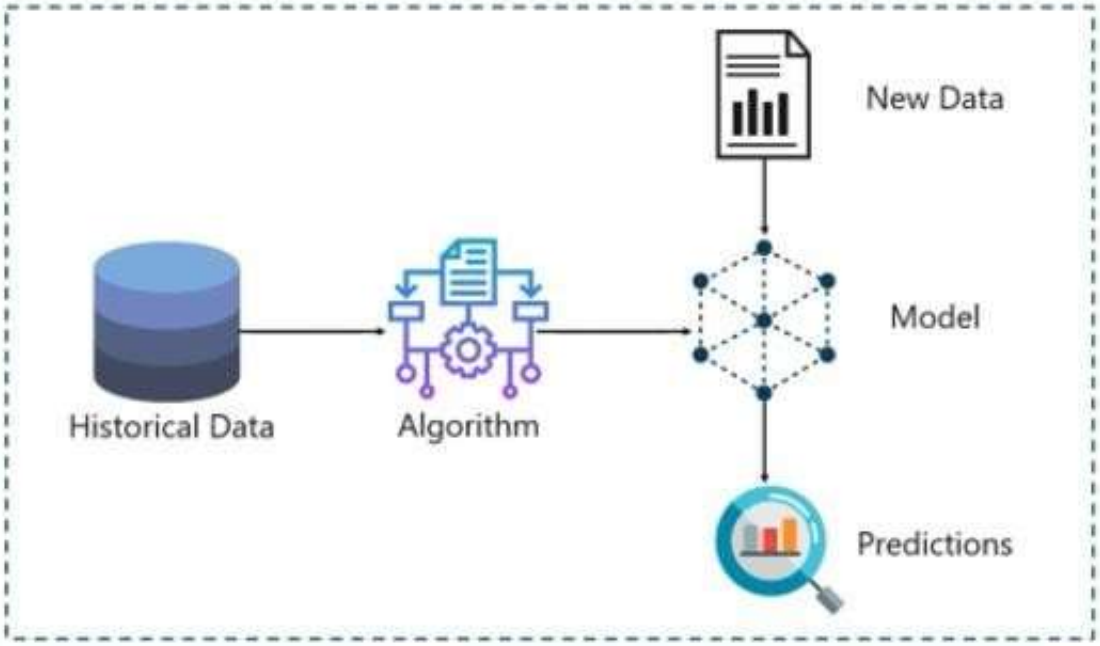
Predictive Modelling

- the process of using known results **to create, process, and validate** a model that can be **used to forecast future outcomes** or predictions.
- **a tool used in** predictive analytics, a data mining technique that attempts to answer the question "what might possibly happen in the future?“.
- Mostly used predictive modeling techniques are :regression and neural networks
  - For example, predictive modeling could help identify customers who are likely to purchase new product over the next 90 days.
- Companies can use predictive modeling to forecast events, customer behavior, as well as financial, economic, and market risks.

- Predictive modeling is a process that uses data and statistics to predict outcomes with data models.
  - These models can be used to predict anything from sports outcomes and TV ratings to technological advances and corporate earnings.
- Predictive modeling is also often referred to as:
  - Predictive analytics
  - Predictive analysis
  - Machine learning

What Is Predictive Analytics?

Predictive analytics is the process of using historical data, machine learning, and artificial intelligence to predict what will happen in the future.



Why predictive modelling

- analysing historical events.
- increase the probability of forecasting events, customer behavior, as well as financial, economic, and market risks

How the data is retrieved?

- rapid migration to digital products has created a **sea of data** that is readily available to businesses.
- **Big data i**s utilized by companies to improve the dynamics of the customer-to-business relationship.
- This vast amount of real-time data is retrieved from sources such as **social media, internet browsing history, cell phone data, and cloud computing platforms**.

Examples

- The bank wants to identify which of its customers are likely to engage in **money laundering(clearing) activities**.
- The bank's customer data, a predictive model is built around the **dollar amount of money transfers** that customers made during a period of time.
- The model is taught to recognize the difference between a money **laundering transaction and a normal transaction**. The optimal outcome from the model should be **a pattern that signals** which customer laundered money and which didn't. If the model perceives that a **pattern of fraud** is emerging for a particular customer, it will create a signal for action, which will be attended to by the bank's fraud prevention unit.

Stages In Predictive Analytics



www.edureka.co/data-science

Process of Predictive Analysis

- Predictive analysis consists of 7 processes as follows:
- **Define project:** Defining the project, scope, objectives and result.
- **Data collection:** Data is collected through data mining providing a complete view of customer interactions.
- Data Cleaning: It is the process of cleaning, inspecting,
- **Data Analysis:** . transforming and modelling the data
  - **Statistics:** This process enables validating the assumptions and testing the statistical models.
- **Predictive Modelling:** Predictive models are generated using statistics and the most optimized model is used for the deployment.
- **Deployment:** The predictive model is deployed to automate the production of everyday decision-making results.
- **Validating the data:** Keep monitoring the model to review performance which ensures expected results.

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Predictive Modelling Tools

- Tools are ability to handle linear and non-linear data relationships.
  - **Neural networks**
  - **decision trees,**
  - **time series data mining, and**
  - **Bayesian analysis**
  - **big data**
    - **Ordinary Least Squares**
    - **Generalized Linear Models (GLM)**
    - **Logistic Regression**
    - **Random Forests**
    - **Decision Trees**
    - **Neural Networks**
    - **Multivariate Adaptive Regression Splines (MARS)**

What are the Biggest Challenges of Predictive Modeling?

- predictive modeling presents a number of challenges in practice.
- These challenges include:
  - **Sufficiently large and comprehensive datasets**
  - **Adaptability of models to new problems**
  - **Data organization and hygiene**
  - **Data privacy and security**

Need of Predictive Analysis

- **Understanding customer behavior:** Predictive analysis uses data mining feature which extracts attributes and behavior of customers. It also finds out the interests of the customers so that business can learn to represent those products which can increase the probability or likelihood of buying.
- **Gain competition in the market:** With predictive analysis, businesses or companies can make their way to grow fast and stand out as a competition to other businesses by finding out their weakness and strengths.
- **Learn new opportunities to increase revenue:** Companies can create new offers or discounts based on the pattern of the customers providing an increase in revenue.
- **Find areas of weakening:** Using these methods, companies can gain back their lost customers by finding out the past actions taken by the company which customers didn't like.

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Applications of Predictive Analysis

- **Health care:** Predictive analysis can be used to determine the history of patient and thus, determining the risks.
- **Financial modelling:** Financial modelling is another aspect where predictive analysis plays a major role in finding out the trending stocks helping the business in decision making process.
- **Customer Relationship Management:** Predictive analysis helps firms in creating marketing campaigns and customer services based on the analysis produced by the predictive algorithms.
- **Risk Analysis:** While forecasting the campaigns, predictive analysis can show an estimation of profit and helps in evaluating the risks too.

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